COL778 A2

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1. Algorithm (high level)

1. Load environment and expert policy.
2. Repeat for each training iteration t:
   - Define mixture probability β_t (probability of querying the expert).
   - Collect N trajectories running a *mixture* policy: with probability β_t use expert; otherwise use learned policy.
   - For every collected state, query the expert to obtain the "ground-truth" action and store (s, a^*) in the replay buffer.
   - Sample minibatches from the replay buffer and minimize mean-squared error between predicted and expert actions (supervised regression).
   - Periodically evaluate the learned policy (run deterministic policy), JIT-save the latest model, and keep the best model according to mean episodic return.

2. Beta schedule (mixing)

The implementation uses a decaying schedule with a lower bound:

$$\beta_t = \max(\beta_{\min}, \alpha^t), \tag{1}$$

with $\beta_{\min} = 0.1$ by default, $\alpha = 0.9$

3. Model, optimizer and loss

- **Actor architecture:** a feed-forward MLP taking observation vector and outputting continuous action vector.
- **Loss:** Mean-squared error.
- **Optimizer:** Adam with configurable learning rate.
- **Device:** model is moved to the configured device (GPU/CPU).

4. Training loop and replay buffer

- Trajectories per iteration: `num_trajs`.
- Maximum trajectory length: `max_length`.
- Replay buffer: capacity 1,000,000 transitions.
- Per-iteration supervised updates: sample minibatches and perform gradient steps.

5. Hyperparameters (selected)

Parameter	Hopper-v4	Ant-v4
batch_size	512	512
max_length	1000	1000
num_trajs	10	10
num_iteration (SGD steps per iter)	500	500
learning_rate	1×10^{-3}	1×10^{-3}
n_layers	2	3
layer_size	64	64
beta_min	0.1	0.1
replay buffer capacity	1,000,000	

Table 1: Primary hyperparameters used for experiments.

6. Evaluation metrics and model selection

- **Primary metric:** Mean episodic return.
- **Secondary metrics:** Mean supervised training loss.
- **Checkpointing:** Best evaluation model retained.

7. Author Contributions

This project was completed with equal contribution from both students. Each student was involved in the design, implementation, experimentation, analysis of results, and preparation of the report. The work distribution was balanced throughout all stages of the project (50–50 contribution).